

NeuroBridge-S4 Dataset Selection and Analysis

Real proxy data, no synthetic subjects, and a modular Control Ladder for HRP methodology review

Dataset Strategy

The challenge asks for methods that work without actual Artemis II data during the submission period. NeuroBridge-S4 uses real public proxy datasets to demonstrate the analysis logic while avoiding synthetic subjects. Each dataset has a specific role in the Control Ladder; no single dataset is treated as a perfect Artemis substitute.

Dataset Name	Primary Role	Strength	Limitation	Use in v4 package
NHANES	Population-scale reference and health-filtered cohort.	Large, public, biomarker/exam variables, real participants.	Cross-sectional; not an astronaut analog.	Executable notebook demo: reference, pseudo-crew, deviation, domains, BACI.
PhysioNet / wearable cohorts	Longitudinal sleep, ECG/HRV, actigraphy, stress/recovery.	Time dynamics; closer to ARChER.	Dataset-specific access and variable heterogeneity.	Defined extension module; notebook structure supports integration.
NASA Life Sciences Portal / UTMB bedrest	Spaceflight analog exposure/recovery.	Small-N and mission-relevant constraints.	Availability and harmonization vary.	Highest-priority extension for Artemis transfer.
ImmPort	Immune study reference.	Immune panels, clinical immunology context.	Registration and study selection required.	Immune Biomarkers extension and stress-immune interpretation.

Why NHANES Remains Useful

NHANES is not used to simulate spaceflight. It is used to demonstrate reference-space construction, cohort filtering, real n=4 pseudo-crew selection, biomarker deviation scoring, biological domain compression, BACI, and graph-informed profile generation. This is the foundation layer; longitudinal and analog datasets become higher Control Ladder levels in full implementation.

Terminology Upgrade

The v4 package avoids overclaiming by replacing “astronaut-like cohort” with “health-filtered terrestrial reference approximation” when NHANES is used alone. This preserves scientific credibility while still giving reviewers a real, reproducible reference cohort.

No Synthetic Subject Policy

The pseudo-crew contains four real public dataset participants selected from the health-filtered reference cohort. Synthetic data may be useful for unit testing, but it should not be the main evidence for a human research methodology. Real proxy data preserve missingness, noise, measurement heterogeneity, and baseline imbalance - exactly the problems the method must handle.

Dataset-to-Method Role Matrix

Method Component	Reference	Observable / Physiological	Behavior / Activity	Immune
Reference distributions	Yes	Partial	Partial	Immune-specific
n=4 pseudo-crew demonstration	Yes	Possible	Possible	Possible
Within-subject time dynamics	No	Yes	Yes	Study-dependent
Sleep/activity layer	No	Yes	Possible	No
Immune-stress layer	Limited	No/limited	Possible	Yes
Recovery trajectory	No	Yes	Yes	Study-dependent

Baseline Imbalance and Missingness

Small samples can be distorted by baseline differences. NeuroBridge-S4 therefore reports filtering decisions, baseline profile differences, missingness patterns, and domain-level completeness. In longitudinal Artemis use, missingness is modeled as both data quality issue and possible operational signal: missed sample, sensor dropout, delayed survey, or incomplete sleep data can indicate mission workload or behavioral burden.

What a 10/10 Demonstration Should Show

What to show	What it means
Reference-space construction	Large real datasets can create interpretable context for rare individuals.
Real n=4 pseudo-crew	The method handles small sample constraints without synthetic people.
Domain compression	Biological interpretation is system-level, not variable spam.
Graph separation	Conceptual knowledge, observed data structure, and decision logic are kept distinct.
BACI sensitivity	Coherence ranking is tested against threshold choices.
Ablation	Each analytic layer adds value beyond raw biomarker z-scores.

Dataset Conclusion

The selected data strategy is intentionally modular. NHANES provides the reproducible foundation. Longitudinal wearable/sleep/stress datasets add ARCHeR-like dynamics. NASA bedrest and ImmPort increase relevance to analog exposure and immune biomarkers. This layered design is more credible than claiming any single public dataset fully reproduces Artemis II.